Instructor key: Paper 003

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Attempt the quiz first. The study guide supplies the reading path. A correct label without its calculation, contract, or evidence limit earns at most 2 of 4 points. Equivalent precise reasoning is acceptable.

Level 0

Q01. Finding: the executable phrase predicate matched. Hold: the policy changed whether or when the proposal could proceed. Acceptance: the route checker found the witness valid under its contract. A permanent block of this eligible, independently valid proposal is false suppression. A temporary hold released before submission is not a permanent block under the question's definition; its delay should be measured separately.

Q02. Both lower-cased sentences contain the same phrase and trigger the same AM-FSP-2 detector. This establishes the phrase predicate's behavior on these two strings. It supplies no semantic verdict about either proposition. The second string is a valid negation of the targeted standard-transform claim under the manuscript's model assumptions, illustrating a dangerous near miss.

Q03. In route I, adding `(a,a)` violates irreflexivity. In L, the same edge is permitted. Bind the input schema, route and applicability, checker version/digest, detector predicate/version, defect code, intended action, positive defect reproduction, independently checked valid controls, fixture digests, and discovery provenance. Admission to I also needs valid controls within I, such as an allowed non-loop edge, relabelings, and a harmless mention of a loop if prose is in scope. A valid L proposal establishes the route boundary; it is not a negative control under I's different validity rule.

Level 1

Q04. The baseline/Appendix A table is:

Inputs	Exact decision	Refuses?
Neither	TOLERATED	No
Pattern only	DIAGNOSTIC_ONLY	No
Structural defect only, LOW/MEDIUM	DIAGNOSTIC_ONLY	No
Structural defect only, HIGH/CRITICAL	DEFENSIVE_LOCKDOWN	Yes
Both, LOW/MEDIUM	DEFENSIVE_LOCKDOWN	Yes
Both, HIGH/CRITICAL	QUARANTINED	Yes

Signal 2 is present here only for a positive defect count. The table evaluates supplied inputs; it does not establish their truth.

Q05. With MEDIUM count zero and a match, the baseline returns DIAGNOSTIC_ONLY; the package returns QUARANTINED. With MEDIUM count one and a match, the baseline returns DEFENSIVE_LOCKDOWN; the package returns QUARANTINED. The differences are Signal 2 presence and the label for low/medium dual-signal refusal. Both tables permit high/critical Signal 2 alone to refuse, so neither is strict conjunction. Do not replace them with the separately corrected historical gate in Appendix A.5.

Q06. Under the HIGH-count derivation, `m=0` yields TOLERATED and every positive `m` yields QUARANTINED. A diagnostic-only match is unreachable. With Signal 2 always absent, zero yields TOLERATED and a positive count yields DIAGNOSTIC_ONLY; nothing refuses. In both cases the decision is still a function of `m` alone. Separate names and selectable test inputs prove neither separate derivation nor accuracy in production. Checks of the same proposal can legitimately be statistically correlated.

Level 2

Q07. The decision is NO_NATIVE_PORT. Fuel becomes 11. The batch rolls back to empty coverage, no witness, and INTAKE; the new receipt/digest records the attempt. There is no terminal certificate. No judgment about the witness's mathematical correctness has occurred.

Q08. The appended move gives REJECT_MOVE_AFTER_VERIFY; fuel becomes 11 and the batch rolls back. The canonical host charges an empty batch and records ACCEPTED; the package charges it and records REJECT_EMPTY_BATCH. With fuel 12, the canonical host spends its twelfth unit and records exhaustion on the next attempted batch. The package enters TERMINAL when the last unit is spent. Neither can issue a route certificate through its missing port.

Q09. Fuel bounds the number of charged processed batches by the initial fuel. It does not bound attempted calls after TERMINAL. At the baseline these calls still append receipts; batch size and serialization also have no bound from fuel alone. A global resource contract needs maximum moves/bytes per batch, call/receipt quotas, retention limits, and time or execution-cost limits. Identifying these missing limits does not imply an accepted semantic transition.

Level 3

Q10. The package checks digest-chain consistency and forbids certificate labels; it does not validate fuel progression or prescribed arm scripts. An editor can supply a matching recomputed chain, and an unchecked fuel edit may not even alter fields covered by that chain. An unrecomputed edit to a covered event or digest fails. The canonical verifier additionally checks fuel, matrix/schema, and scripted decisions, so the same arbitrary fuel rewrite fails there. Neither unkeyed chain proves who produced the record. A trusted external anchor

detects a complete rewrite relative to that anchor; route verification remains a separate semantic check.

Q11. The records identify different runner behaviors and cannot be pooled as if they came from one randomized frozen experiment. Historical certificate reproduction requires the original runner/checker bytes, route adapters, task inputs, proposal bytes, parameters, and receipt replay. A current causal claim also requires a validated checker, actual allocation record, fixed policy/budgets, independently reconstructed outcomes, and the declared analysis. The historical pilot is descriptive evidence with its own limits.

Q12.

Pin	What it identifies	What it does not establish
e11c07ce90318309f058c8097	Source-tree state used for this review	Execution of this review's modified bytes
52cb5457f6c485ba548cfd292	Default-branch baseline for the main audit	Behavior of later default-branch code
30609de2c3928238715375a9	Source audited in Appendix A	Activation or effectiveness after the audit
fdc0035a81c45e10b900fc917	Historical source revision	Fresh execution or current integration

A fresh receipt adds exact tested file hashes, commands, tool versions, outcome counts, exit statuses, and skipped/not-run scope. A default-branch assertion requires inspecting and testing that branch's current bytes.

Level 4

Q13. Positive aggregation gives `theta=3/2` for the first list, so `H=(9/4)/(4+9/4)=9/25<1/2`: no activation. For `[2]`, `H=4/8=1/2`: activation. The threshold is `theta>=2`. The definition discards negative entries instead of allowing them to cancel positives.

Q14. At most `3m/4` can come from `m` positive entries. Activation requires `3m/4>=2`, so `m>=ceil(8/3)=3`; three entries of `3/4` suffice. This counts entries. Identity and provenance conditions are needed to count distinct independent witnesses. Two copied unit entries reach the sum threshold but provide only one underlying source. They also lie outside the specified `3/4` cap, so that example concerns the uncapped/unit-capped gate.

Q15. For `0<t<1`, `theta^n/(K^n+theta^n)>=t` is equivalent to `theta^n(1-t)>=tK^n`, hence `theta>=K(t/(1-t))^(1/n)`. At zero input the inequality is false for this

open interval. At `t=0`, the gate activates even with no signals. At `t=1`, no finite sum activates for positive `K,n`. The proof allows positive real `n`, while the exact helper uses a positive integer exponent. `True` and `Fraction(3,2)` produce LemmaInputError.

Level 5

Q16. Since `P intersect S` is empty, the screen admits the rule. Since `E intersect S={a}`, the executed rule fires on a valid self member. The singleton panel `{a}` is already the minimal nonempty counterexample. If bit 0 denotes `a`, then `P=2,E=3,S=1` replay this case.

Q17. If `E implies P`, an admitted rule with no P-hit on S can have no E-hit there. Conversely, if some `x` satisfies E but not P, then `S={x}` admits the rule and is hit by its execution. Thus soundness for every finite panel is equivalent to `E implies P`. Equality is unnecessary for soundness alone: screening may be more conservative than execution.

Q18. If `P implies E`, any screened hit is an executed hit, so a rule that would not fire on S is not rejected. If the implication fails at `x`, the singleton S rejects a rule whose execution is harmless on S. Both directions together require equality. In the supplied `E=P={b},S={a}` case the screen faithfully tests execution on a, but b is absent from the panel. If b is independently valid, the admitted rule can still falsely suppress b. Predicate agreement guarantees panel-relative behavior, not universal coverage or a measured rate.

Level 6

Q19. The three off-diagonal entries of `SHS` are all -1. The signed cycle product is `(-1)(-1)(+1)=+1`, consistent with the gauge. With all positive bonds it is `(-1)^3=-1`, obstructing a diagonal sign gauge. The underlying graph is a triangle in both cases; the signed cycle differs. The standard local particle-hole constraint requires `epsilon_0 epsilon_1=epsilon_1 epsilon_2=epsilon_2 epsilon_0=-1`. Multiplying makes `1=-1`, so it fails on this odd cycle regardless of the bond signs in the real-hopping setting.

Q20. Under the declared common real hopping and equal spin chemical potentials, charge-channel decoupling gives equal real spin-sector determinants d for each auxiliary-field configuration. Their product `d*d>=0` needs no bipartite graph. It can be zero, so "strictly positive" would overstate the argument. This determinant factorization does not establish the local particle-hole sign assignment. Equal populations alone do not specify identical one-body spin dynamics or the required decoupling and chemical potentials.

Q21. A restricted sign-free model can be valid and consistent with worst-case hardness. The generic polynomial-time claim needs a precise encoding, class, accuracy/cost contract, and algorithm covering the NP-hard reduction; under the stated hardness result it would have the corresponding polynomial complexity

consequences. Hardness does not prove P differs from NP or forbid all specialized methods. Majorana notation alone proves no positivity; the published sufficient conditions must be checked on the actual model. Meron cancellation likewise needs its cluster pairing and equal-weight conditions. These checks are unavailable to a phrase detector.

Level 7

Q22. Effectiveness is `3/10=30%`; none of the seven other allocated runs is removed. Report its outcome category separately. False suppression is `2/20=1/10=10%`. The 100 blocked invalid proposals are outside the conditional denominator and cannot reduce the rate to `2/120`. A route-checker error invalidates the affected semantic analysis rather than becoming a model defect.

Q23. Solve `(1-p)^n=0.05`: the upper bound is `1-0.05^(1/n)`. For eight zero-failure opportunities it is about 0.3123, or 31.23%. For a strict upper bound below 1%, the equivalent test is `(99/100)^n<1/20`. This is false at 298 and true at 299. Tagged copies cannot establish independent sampling; even 40 distinct strings need not be independent trials. Three separate 95% bounds do not automatically give simultaneous 95% coverage. For example, Bonferroni uses `alpha=0.05/3` per route, requiring 408 zero-failure independent opportunities per route for the same strict 1% bound. Dependence handling and any repeated looks must also be prespecified.

Q24. A sufficient design declares:

- Validated route adapters and defect predicates, an applicable independent checker, and a frozen detector/admission bank.
- Instance-level separation of discovery, admission, pilot, and confirmatory tasks, with reserved families/models and route/difficulty strata.
- Recorded random allocation/order within fixture/model blocks; analysis at the actual dependence unit, including shared trials from one underlying fixture.
- Identical schemas, task inputs, stopping budgets, and feedback opportunities for D and R, with prespecified feedback content and resource accounting.
- Independently verified run success over all allocated runs; safe shadow evaluation of valid eligible blocked proposals with explicit denominators.
- A positive lower 95% endpoint for the principal held-out D-minus-R comparison, the route-specific false-suppression gate, zero false terminal certificates, and declared multiplicity/interval handling.
- Source/detector/checker/task digests and raw authorized proposal/assignment records; separate refused, unknown, timeout, and exhausted outcomes.
- Stops for budget, false terminal, checker defect, or source/custody compromise; a new source-bound run when a detector or checker changes.

Randomization and finite fixtures alone do not establish power or independence. The final analysis and sample-size plan must be frozen before confirmatory

data. An unavailable adapter leaves effectiveness unassessed. Arm A evaluates solution-bearing assistance and cannot support unassisted performance claims.

Replay the numerical answers

From the package root, this standard-library worksheet checks the nontrivial fractions and confidence thresholds without requiring a provider or Rust:

```
import sys
from fractions import Fraction as F
sys.path.insert(0, "code")
from detector_lemmas import admitted, fires_on_self, hill_activates
assert F(9, 4) / (4 + F(9, 4)) == F(9, 25)
assert not hill_activates([1, F(1, 2), -7], 2, 2, F(1, 2))
assert hill_activates([2], 2, 2, F(1, 2))
assert admitted(2, 1) and fires_on_self(3, 1)
assert F(99, 100) ** 298 >= F(1, 20)
assert F(99, 100) ** 299 < F(1, 20)
assert F(99, 100) ** 407 >= F(1, 60)
assert F(99, 100) ** 408 < F(1, 60)
```

These calculations check the declared exercise inputs. The manuscript proofs carry the general mathematical arguments under their assumptions; this worksheet supplies no empirical safety guarantee.